

# Packet Tracer Lab 11 - Identify MAC and IP Addresses

## Objectives

**Part 1: Gather PDU Information for Local Network Communication**

**Part 2: Gather PDU Information for Remote Network Communication**

## Background

This activity is optimized for viewing PDUs. The devices are already configured. You will gather PDU information in simulation mode and answer a series of questions about the data you collect.

## Instructions

### Part 1: Gather PDU Information for Local Network Communication

**Note:** Review the Reflection Questions in Part 3 before proceeding with Part 1. It will give you an idea of the type of information you will need to gather. Gather PDU information as a packet travels from 172.16.31.5 to 172.16.31.2.

- Click **172.16.31.5** and open the **Command Prompt**.
- Enter the **ping 172.16.31.2** command.
- Switch to simulation mode and repeat the **ping 172.16.31.2** command. A PDU appears next to **172.16.31.5**.
- Click the PDU and note the following information from the **OSI Model** and **Outbound PDU Layer** tabs:
  - Destination MAC Address: **000C:85CC:1DA7**
  - Source MAC Address: **00D0:D311:C788**
  - Source IP Address: **172.16.31.5**
  - Destination IP Address: **172.16.31.2**
  - At Device: **172.16.31.5**
- Click **Capture / Forward (the right arrow followed by a vertical bar)** to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered into a spreadsheet using a format like the table shown below:

### Example Spreadsheet Format

| At Device   | Dest. MAC      | Src MAC        | Src IPv4    | Dest IPv4   | Type     |
|-------------|----------------|----------------|-------------|-------------|----------|
| 172.16.31.5 | 000C:85CC:1DA7 | 00D0:D311:C788 | 172.16.31.5 | 172.16.31.2 | Ethernet |
| Switch1     | 000C:85CC:1DA7 | 00D0:D311:C788 | 172.16.31.5 | 172.16.31.2 | Ethernet |
| Hub         | 000C:85CC:1DA7 | 00D0:D311:C788 | 172.16.31.5 | 172.16.31.2 | Ethernet |
| 172.16.31.2 | 00D0:D311:C788 | 000C:85CC:1DA7 | 172.16.31.2 | 172.16.31.5 | Ethernet |

| At Device   | Dest. MAC      | Src MAC        | Src IPv4    | Dest IPv4   | Type     |
|-------------|----------------|----------------|-------------|-------------|----------|
| 172.16.31.2 | N/A            | N/A            | 172.16.31.2 | 172.16.31.3 | IP       |
| 172.16.31.2 | 0060.7036.2849 | 000C:85CC:1DA7 | 172.16.31.2 | 172.16.31.3 | Ethernet |
| Hub         | 0060.7036.2849 | 000C:85CC:1DA7 | 172.16.31.2 | 172.16.31.3 | Ethernet |
| Switch1-x   | 0060.7036.2849 | 000C:85CC:1DA7 | 172.16.31.2 | 172.16.31.3 | Ethernet |
| 172.16.31.3 | 000C:85CC:1DA7 | 0060.7036.2849 | 172.16.31.3 | 172.16.31.2 | Ethernet |
| 172.16.31.4 | N/A            | N/A            | 172.16.31.4 | 172.16.31.5 | IP       |
| 172.16.31.4 | 00D0.D311.C788 | 000C.CF0B.BC80 | 172.16.31.4 | 172.16.31.5 | Ethernet |
| Switch1     | 00D0.D311.C788 | 000C.CF0B.BC80 | 172.16.31.4 | 172.16.31.5 | Ethernet |
| 172.16.31.5 | 000C.CF0B.BC80 | 00D0.D311.C788 | 172.16.31.5 | 172.16.31.4 | Ethernet |

### Step 2: Gather additional PDU information from other pings.

Repeat the process in Step 1 and gather the information for the following tests:

· Ping 172.16.31.2 from 172.16.31.3

· Ping 172.16.31.4 from 172.16.31.5

Return to Realtime mode.

## Part 2: Gather PDU Information for Remote Network Communication

In order to communicate with remote networks, a gateway device is necessary. Study the process that takes place to communicate with devices on the remote network. Pay close attention to the MAC addresses used.

### Step 1: Gather PDU information as a packet travels from 172.16.31.5 to 10.10.10.2.

- Click **172.16.31.5** and open the **Command Prompt**.
- Enter the **ping 10.10.10.2** command.
- Switch to simulation mode and repeat the **ping 10.10.10.2** command. A PDU appears next to **172.16.31.5**.
- Click the PDU and note the following information from the **Outbound PDU Layer** tab:
  - Destination MAC Address: 00D0:BA8E:741A, Source MAC Address: 00D0:D311:C788, Source IP Address: 172.16.31.5, Destination IP Address: 10.10.10.2, At Device: 172.16.31.5

What device has the destination MAC that is shown?

- Click **Capture / Forward (the right arrow followed by a vertical bar)** to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered from pinging 172.16.31.5 to 10.10.10.2 into a spreadsheet using a format like the sample table shown below:

**PDU Data Table**

| At Device   | Dest. MAC 1<br>Ethernet Header | MAC 2 Address | Src MAC 1<br>Ethernet Header | Src IPv4<br>IP Header | Dest IPv4<br>IP Header | Packet<br>Direction | Type             | Highest<br>OSI Layer |
|-------------|--------------------------------|---------------|------------------------------|-----------------------|------------------------|---------------------|------------------|----------------------|
| 172.16.31.5 | 000C.85CC.1DA7                 |               | 00D0:D311:C788               | 172.16.31.5           | 172.16.31.2            | outbound            | Gigabit Ethernet | 3 network            |
| Switch1     | 000C.85CC.1DA7                 |               | 00D0:D311:C788               | 172.16.31.5           | 172.16.31.2            | Outbound            | Gigabit Ethernet | 2 data link          |
| Hub         | 000C.85CC.1DA7                 |               | 00D0:D311:C788               | 172.16.31.5           | 172.16.31.2            | Outbound            | Fast Ethernet    | 1 physical           |
| 172.16.31.2 | 00D0:D311:C788                 |               | 000C.85CC.1DA7               | 172.16.31.2           | 172.16.31.5            | Outbound            | Gigabit Ethernet | 3 network            |
| 172.16.31.3 | 000C.85CC.1DA7                 |               | 00D0:D311:C788               | 172.16.31.5           | 172.16.31.2            | Inbound             | Gigabit Ethernet | 0                    |
| Hub         | 00D0:D311:C788                 |               | 000C.85CC.1DA7               | 172.16.31.2           | 172.16.31.5            | Outbound, ACK       | Fast Ethernet    | 1 physical           |
| Switch1     | 00D0:D311:C788                 |               | 000C.85CC.1DA7               | 172.16.31.2           | 172.16.31.5            | Outbound, ACK       | Gigabit Ethernet | 2 data link          |
| 172.16.31.3 | 00D0:D311:C788                 |               | 000C.85CC.1DA7               | 172.16.31.2           | 172.16.31.5            | Inbound             | Gigabit Ethernet | 2 data link          |
| 172.16.31.5 | 00D0:D311:C788                 |               | 000C.85CC.1DA7               | 172.16.31.2           | 172.16.31.5            | Inbound, ACK        | Gigabit Ethernet | 3 network            |
| 172.16.31.3 | 00D0.BA8E.741A                 |               | 0060.7036.2849               | 172.16.31.3           | 10.10.10.3             | outbound            | Gigabit Ethernet | 3 network            |

| At Device    | Dest. MAC 1 Ethernet Header | MAC 2 Address  | Src MAC 1 Ethernet Header | Src IPv4 IP Header | Dest IPv4 IP Header | Packet Direction | Type             | Highest OSI Layer |
|--------------|-----------------------------|----------------|---------------------------|--------------------|---------------------|------------------|------------------|-------------------|
| Hub          | 00D0.BA8E.741A              |                | 0060.7036.2849            | 172.16.31.3        | 10.10.10.3          | Outbound         | Fast Ethernet    | 1 physical        |
| Switch1      | 00D0.BA8E.741A              |                | 0060.7036.2849            | 172.16.31.3        | 10.10.10.3          | Outbound         | Gigabit Ethernet | 2 data link       |
| 172.16.31.2  | 00D0.BA8E.741A              |                | 0060.7036.2849            | 172.16.31.3        | 10.10.10.3          | Inbound          | Gigabit Ethernet | 2 data link       |
| Router       | 00D0.BA8E.741A              |                | 0060.7036.2849            | 172.16.31.3        | 10.10.10.3          | Outbound         | Gigabit Ethernet | 3 network         |
| Router       | 0060.4706.572B              |                | 00D0.588C.2401            | 172.16.31.3        | 10.10.10.3          | Outbound         | Gigabit Ethernet | 3 network         |
| Switch0      | 0060.4706.572B              |                | 00D0.588C.2401            | 172.16.31.3        | 10.10.10.3          | Outbound         | Gigabit Ethernet | 2 data link       |
| Access Point | 0060.4706.572B              |                | 00D0.588C.2401            | 172.16.31.3        | 10.10.10.3          | Inbound          | Gigabit Ethernet | 1 physical        |
| Access Point | 0060.4706.572B              | 0050.0FAB.6C82 | 00D0.588C.2401            | 172.16.31.3        | 10.10.10.3          | Outbound         | 802.11 Wireless  | 1 physical        |
| 10.10.10.3   | 0050.0FAB.6C82              | 0060.4706.572B | 00D0.588C.2401            | 10.10.10.3         | 172.16.31.3         | Outbound         | 802.11 Wireless  | 3 network         |
| 10.10.10.2   | 0060.4706.572B              | 0050.0FAB.6C82 | 00D0.588C.2401            | 172.16.31.3        | 10.10.10.3          | Inbound          | 802.11 Wireless  | 1 physical        |
| 10.10.10.3   | 0050.0FAB.6C82              | 0060.4706.572B | 00D0.588C.2401            | 10.10.10.3         | 172.16.31.3         | Outbound         | 802.11 Wireless  | 1 physical        |

| At Device    | Dest. MAC 1 Ethernet Header | MAC 2 Address  | Src MAC 1 Ethernet Header | Src IPv4 IP Header | Dest IPv4 IP Header | Packet Direction | Type             | Highest OSI Layer |
|--------------|-----------------------------|----------------|---------------------------|--------------------|---------------------|------------------|------------------|-------------------|
| Access Point | 00D0.588C.2401              |                | 0060.4706.572B            | 10.10.10.3         | 172.16.31.3         | Outbound, ACK    | Ethernet         | 1 physical        |
| Switch0      | 00D0.588C.2401              |                | 0060.4706.572B            | 10.10.10.3         | 172.16.31.3         | Outbound, ACK    | Ethernet         | 2 data link       |
| Router       | 0060.7036.2849              |                | 00D0.BA8E.741A            | 10.10.10.3         | 172.16.31.3         | Outbound, ACK    | Ethernet         | 3 network         |
| Access Point | 00D0.588C.2401              | 0050.0FAB.6C82 | 0060.4706.572B            | 10.10.10.3         | 172.16.31.3         | Outbound         | Ethernet         | 1 physical        |
| 10.10.10.3   | 00D0:588C:2401              | 0050.0FAB.6C82 | 0060.4706.572B            | 10.10.10.2         | 172.16.31.3         | Inbound          | 802.11 Wireless  | 1 physical        |
| 10.10.10.2   | 0060.4706.572B              | 0050.0FAB.6C82 | 00D0.588C.2401            | 172.16.31.3        | 10.10.10.3          | Inbound          | 802.11 Wireless  | 1 physical        |
| Switch1      | 0060.7036.2849              |                | 00D0.BA8E.741A            | 10.10.10.3         | 172.16.31.3         | Outbound, ACK    | Gigabit Ethernet | 2 data link       |
| Hub          | 0060.7036.2849              |                | 00D0.BA8E.741A            | 10.10.10.3         | 172.16.31.3         | Outbound, ACK    | Fast Ethernet    | 1 physical        |
| 172.16.31.2  | 0060.7036.2849              |                | 00D0.BA8E.741A            | 10.10.10.3         | 172.16.31.3         | Inbound          | Gigabit Ethernet | 3 network         |
| 172.16.31.3  | 0060.7036.2849              |                | 00D0.BA8E.741A            | 10.10.10.3         | 172.16.31.3         | Inbound, ACK     | Gigabit Ethernet | 3 network         |

## Reflection Questions

Answer the following questions regarding the captured data:

1. Were there different types of cables/media used to connect devices? **Wired Ethernet and wireless**
2. Did the cables change the handling of the PDU in any way? **No. The transmission media (Ethernet cable or wireless) did not change the IP packet. Different media only changed the Layer 2 frame format used to carry the packet.**
3. Did the **Hub** lose any of the information that it received? **No, but it sent information to a device that dropped the PDU b/c the device was not the destination.**
4. What does the **Hub** do with MAC addresses and IP addresses? **A hub does not examine or modify MAC or IP addresses. It simply repeats incoming electrical signals out every other port.**
5. Did the wireless **Access Point** do anything with the information given to it? **No. the IP packet stays exactly the same.**
6. Was any MAC or IP address lost during the wireless transfer? **No**
7. What was the highest OSI layer that the **Hub** and **Access Point** used? **The Hub reached layer 1 physical and the AP reached layer 2 data link**
8. Did the **Hub** or **Access Point** ever replicate a PDU that was rejected with a red "X"? **No, there was never an ACK sent back from the node that rejected the PDU**
9. When examining the **PDU Details** tab, which MAC address appeared first, the source or the destination? **Both the AP and the HUB have the destination first, except when the AP has 4 MAC address spaces.**
10. Why would the MAC addresses appear in this order? **Because the purpose is delivery to the destination address so it's first**
11. Was there a pattern to the MAC addressing in the simulation? **Yes, when the PDU is traveling to the destination, then the destination is first, but once at the destination, the source and destination flip and the ACK is then sent to the initial source (now the destination).**
12. Did the switches ever replicate a PDU that was rejected with a red "X"? **No, there was never an ACK sent back from the node that rejected the PDU. The ACK that was sent from the target node was replicated to all connected devices.**
13. Every time that the PDU was sent between the 10 network and the 172 network, there was a point where the MAC addresses suddenly changed. Where did that occur? **It changed at the router, where the MAC address changes at every hop.**
14. Which device uses MAC addresses that start with 00D0:BA? **172.16.31.3**
15. What devices did the other MAC addresses belong to? **10.10.10.3**

16. Did the sending and receiving IPv4 addresses change fields in any of the PDUs? **Yes, when the packet was received by the target device, the ACK as the destination and source IP flipped.**
17. When you follow the reply to a ping, sometimes called a *pong*, do you see the sending and receiving IPv4 addresses switch? **Yes, the pong "ACK" is reversed from the initial request.**
18. What is the pattern to the IPv4 addressing used in this simulation? **Each port on the router belongs to one network. One port leads to the 172 thru switch 0 and the other leads to the 10.10 thru switch 1. The route connects together.**
19. Why do different IP networks need to be assigned to different ports of a router? **Each router interface belongs to a different IP network. Routers separate broadcast domains and provide routing between different IP subnets.**
20. If this simulation was configured with IPv6 instead of IPv4, what would be different? **The communication process would be the same, but IPv6 addresses would replace IPv4 addresses, ARP would be replaced by Neighbor Discovery (NDP), and ICMPv6 would be used instead of ICMP for address resolution and other functions.**